Skill Activity 1 – Identify the ML Solution

A college wants to develop a system that predicts whether a student is likely to successfully complete a course based on attendance, internal marks, assignment performance, previous academic performance, and practical performance.

1. What is the problem?
   The problem is to predict whether a student will successfully complete a course based on information about their academic performance and participation and to develop an ML system that predicts that outcome

2. What are the input features?
   Attendance
   Internal marks
   Assignment performance
   Previous academic performance
   Practical performance

3. What is the target variable?
   The course completion status, such as Complete or Not Complete.

4. Is this classification or regression?
   It is a binary classification problem because the target has two possible categories.

5. Which ML algorithm could be used?
   Logistic Regression can be used. Other suitable algorithms include Decision Tree and Random Forest if there are non-linear relationships between the features in the dataset.

6. What type of data is required?
   Historical data, structured student data containing the input features along with the actual course completion outcome. The data should be:
   - Accurate: Marks and attendance should be recorded correctly:
   - Complete: there shouldn't be excessive missing values
   - Relevant: the features should actually have some relationship with course completion
   - Representative: the dataset should contain students with different performance levels
   - structured
   - sufficiently complete

7. How will the model be evaluated?
   The data can be divided into training and testing sets. The model can be evaluated using:
   - Accuracy: How many predictions were correct overall
   - Precision: Of the students predicted as complete, how many actually completed?
   - Recall: Of all students who actually did not complete, how many did the model correctly identify as at risk (TN case)
   - F1-score: a combination of precision and recall
   - confusion matrix: gives us an understanding of what kind of mistakes the model is making

8. What challenges could occur?
   Possible challenges include:
   - missing data
   - incorrect records
   - class imbalance
   - small datasets
   - overfitting
   - bias
   - changing academic patterns
   - test data leakage.